

KubeSight — Full Repository Audit & Feature Inventory

Branch: `master` (HEAD `650996c4`) · **Date:** 2026-09-25 · **Method:** read-only static audit, 8 parallel sweeps tracing UI → API → service → DB/external. No files were changed and nothing was executed. Anything that depends on a live cluster, VM, Jenkins, Bitbucket, Hermes or store API is marked *uncertain — requires runtime verification*.

Branch note: the Hermes ticket agent (Telegram approvals + 5 MCP ticket tools) exists **only on** `master` — not on `main`, `production` or `test/merge-check`.

Legend: Implemented · Partial · Frontend only · Backend only · Infra/config only · Planned / not implemented · Legacy / superseded

0. Top findings to act on (security / correctness)

- Live API token committed to git.** `connect-hermes-to-kubesight.sh` is tracked and hard-codes a full `ksa_...` token twice (in history since `c50724fb`). Rotate it.
- A "green" CI build may have run nothing.** Every install seeds a `kubesight-mock` runner that is enabled and online, while the real Kubernetes runner is seeded disabled (`migrate_rbac.py` `_seed_builtin_ci_runners`). The mock runner reports success after ~2 s without running anything, and an unpinned stage can land on it.
- Default users with known passwords always seeded:** `admin/admin123`, `operator/operator123`, `viewer/viewer123` — MFA off, onboarding already complete (`seed.py`, `rbac_data.py`).
- API-token routes check no permission.** `routes/api_tokens.py` never checks `api_tokens:manage` (only the sidebar does) and skips the first-login gate. Tokens have no scopes — each carries the owner's full RBAC.
- Kubeconfigs stored as plaintext files** in `data/kubeconfigs/cluster-N.yaml`; Fernet encryption is used for every other stored secret but not these.
- `GET /api/alerts` **sends notifications as a side effect** (`dispatch_firing_alerts`).
- Hermes-generated CI pipelines are auto-saved without review** and overwrite the service's default pipeline (`ci_assist/generator.py` → `accept.auto_accept`), contradicting the package's own "a person must approve" rule.
- Approval gate only covers** `apply.yaml`. Helm install/upgrade and restart/scale/rollback bypass `assert_deploy_allowed`. The typed confirmation phrase for YAML/image/wizard applies is checked only in the frontend.
- Ticketing webhooks accept anything when no inbound secret is configured** (Zoho and Jira).
- The root** `Jenkinsfile` **deploy stage applies 9** `k8s/*.yaml` **files that don't exist** (namespace, backend-deployment, frontend-deployment, ...).

1. Feature map, status and evidence

1.1 Platform / Core

Feature	Status	Evidence / what it proves
Password login (username or email)		<code>LoginPage.jsx</code> → <code>routes/auth.py</code> <code>/login</code> → <code>auth_service.py</code> ; stage machine returns <code>authenticated</code> / <code>password_change</code> / <code>mfa_setup</code> / <code>mfa_challenge</code>
JWT sessions with purpose claim		<code>auth_utils.py</code> : access 8 h, onboarding/mfa 30 min, download 120 s
TOTP MFA (enrolment + challenge)		<code>totp_service.py</code> (pyotp, QR). TOTP secret stored in plaintext
First-login onboarding (temp pwd → change → TOTP)		<code>OnboardingPage.jsx</code> , <code>user_service._provision_temporary_password</code>
Lockout		5 failures → 15-min lock; 3rd lock in 24 h → admin unlock; HTTP 423
Security emails		<code>email_delivery.py</code> ; if SMTP fails the temp password is returned in the API response. Delivery uncertain
Logout		Audit row only; JWT not revoked server-side

Feature	Status	Evidence / what it proves
Users CRUD + admin recovery	✓	UserManagementPage.jsx → routes/users.py ; unlock, reset MFA, resend temp pwd, force reset, lock, enable, permanent delete; last-admin and self-demotion guards
Roles & permissions (~110 keys)	✓	role_service.py , rbac_data.py ; system roles admin, operator, cluster_admin, viewer, hermes-agent
Fine-grained access rules	✓	access_engine.py , AccessRulesEditor.jsx ; cluster → namespace → resource → container → port, allow/deny
Legacy cluster/namespace grants	🗑	UserClusterAccess / UserNamespaceAccess ; only for users without AccessRules
Effective access summary	✓	access_summary.py drives page gating
API tokens (ksa_ , sha256)	●	Works, but no backend permission check and no scopes
Audit log + CSV export	✓	audit.py used everywhere; AuditLogsPage.jsx
Workspace settings	✓	routes/settings.py ; theme per browser
Settings → Security card	●	Display only; no self-service password / MFA change
Secret encryption at rest (Fernet)	✓	secret_encryption.py ; key falls back to JWT key then a dev default
DB bootstrap + migrations	✓ custom	db.create_all() + ~35 idempotent ALTERs in migrate_rbac.py ; silent SQLite fallback outside production
Alembic	🗑	Only .pyc files remain
Mock vs real Kubernetes (K8S_REAL_MODE)	✓	k8s_provider.py , mock_data.py
AUTH_REQUIRED=false kill switch	●	Bypasses every decorator
KUBESIGHT_ENV production mode	✗	Not an app flag — only a CI build-env file variable
TTL cache + cache warmer	✓	ttl_cache.py , cache_warmer.py
Hash routing, coach-mark tours	✓	routeTable.js , authz.js NAV_PAGES , frontend/src/tours
Containers / runtime	●	backend/Dockerfile (kubectl 1.32, helm 3.17); gunicorn pinned to 1 worker (in-process schedulers)
docker-compose	●	Postgres only

1.2 Kubernetes cluster management (existing clusters)

All Kubernetes operations shell out to `kubect1` via `k8s_provider` ; no Python kubernetes client.

Feature	Status	Evidence
Auto-discover kubeconfig contexts	✓	_discovered_clusters_from_k8s
Register by kubeconfig upload/paste	✓	ClusterManagementPage.jsx → POST /api/clusters/custom → cluster_connection_service.py → cluster_store.py
Register manually (token / cert / anonymous)	✓	kubeconfig_builder.py
In-cluster service account	●	k8s_entrypoint.sh
Update / test cluster	✓	PUT /custom/<ref> , POST .../test
Delete cluster	●	Soft delete; kubeconfig file left on disk
Multi-cluster list	✓	GET /api/clusters , 30 s cache, per-user filter
Cluster health	●	healthy / warning only; circuit breaker
Cluster overview	●	Pods, CPU, memory real; workload & storage counts hard-coded 0
Namespaces	●	List only; created only as deploy side effect
Pods	✓	List + metrics, describe, YAML, restart, logs, exec
Deployments	✓	View, describe, YAML, edit/apply (validate → dry-run → diff → apply), rollout restart; scale/rollback only on ApplicationDetailsPage
StatefulSets / DaemonSets	●	View, describe, YAML, rollout restart

Feature	Status	Evidence
ReplicaSets / Jobs / CronJobs	●	View only
Services	●	View only
Ingress / ConfigMaps / Secrets	✓	View + edit/apply; Secret YAML includes base64 data
Events	✓	Only surfaced on ApplicationDetailsPage
Problem pods across namespaces	✓	GET /pod-issues
PVC / PV browsing	✗	
StorageClass	●	List only
Nodes (control-plane vs worker)	●	Roles in dashboard widget; no Nodes page, no cordon/drain
Resource delete	✗	Only via change-bundle delete_deployment
Exec	✓	One-shot sh -c , no TTY
Topology map	✓	topology_service.py , TopologyPage.jsx
K8s version upgrades (Upgrade Center)	●	kubeadm + minikube execute; EKS/AKS/GKE precheck only; jobs in memory. Uncertain

1.3 Cluster provisioning (Cluster Builder)

Feature	Status	Evidence
Build wizard + CRUD with topology validation	✓	Wizard.jsx → routes/cluster_builds.py → cluster_build/service.py
Preflight	✓	cluster_build/preflight.py
Phase machine (base_prep → LB → images → init → CNI → join CPs → join workers → verify → onboard → add-ons → workloads)	✓ code, runtime uncertain	cluster_build/executor.py ; real-VM validation ladder pending
Resume, retry, cancel, scrubbed live logs	✓	single-process assumption
Stacked HA (3 CPs)	✓	
Managed HAProxy + Keepalived + VIP	✓ code, uncertain	cluster_build/lb.py
External LB / manual endpoint	✓	Validation only
Calico (default) / Flannel	✓	Bundled manifests in backend/api/data/cni
Cilium	●	KUBESIGHT_ENABLE_CILIUM ; no bundle
Add-ons: metrics-server, NGINX Ingress, MetalLB	✓	cluster_build/addons
NFS provisioner / StorageClass add-on	✗	
OS adapters Debian/Ubuntu, RHEL/Rocky	✓	RHEL unvalidated on real VMs
Build profiles (internet / mirror / offline / proxy / CA)	✓	
Auto-onboard, kubeconfig download, Open in Clusters	✓	
Day two: add workers	✓ with gap	metrics-server CSR approver not re-rendered → new nodes' metrics/logs likely fail
Remove live node, add CP/LB, teardown, IPAM, kube-vip	✗	
Copy workloads from another cluster	✓	cluster_build/workloads.py
Copied PVC storage plan (NFS fresh / reuse / class / pending)	✓	cluster_build/storage.py
vSphere connections	✓ read-only	edit is ●
SSH credentials / profiles (direct / bastion)	✓	edit is ●
SSH host-key policies strict / TOFU / pinned	●	preapprove() has no route/UI — pinned & strict can't be satisfied

1.4 Monitoring / observability

Feature	Status	Evidence
Pod & node CPU / memory	✓	kubectl top via metrics-server (k8s_metrics.py)
Prometheus / Grafana / Loki	✗	mock data only
Disk metrics	✗	DiskPressure flag only
Network metrics	● fake	random walk in useDashboardSeries.js
Historical metrics	●	client buffer seeded with random values; nothing stored
Dashboard summary	✓	GET /api/dashboard/summary → dashboard_service.py; OpsDashboard.jsx
Logs snapshot / SSE follow	✓	logs_service.py, /logs/stream
Noise filter, search, level filter, download	✓	log_noise.py, LogViewer.jsx
Aggregated / stored logs	✗	
Legacy GET /api/logs	🗑 still used	ApplicationDetailsPage only

1.5 Alerts

Feature	Status	Evidence
Built-in CPU-over-limit alerts	✓	80 % of pod CPU limit
Policy CRUD, enable/disable, catalog	✓	AlertPoliciesPage.jsx → alert_policy_service.py
Metric rules (CPU, mem, restarts, NotReady, CrashLoop, Pending, unavailable replicas)	✓	alert_policy_evaluator.py
PVC % / disk % rules	● misleading	hard-coded values, not real usage
Log-pattern alerts	✓	log_alert_evaluator.py; never auto-resolve
Service-health / automation-failure alerts	✓	service_alert_evaluator.py, automation_alert_service.py
Scheduler (15 s tick)	✓	alert_policy_scheduler.py — also drives builds, automation, bundles, mobile, ticket agent
Metric auto-resolve	✓	
Delivery: SMTP, Slack webhook, generic webhook	✓	alert_routing_service.py
Receivers, groups, repeat interval, delivery log	✓	embedded in Settings → Integrations
Telegram / WhatsApp / Teams / PagerDuty alerts	✗	
Resolved notifications, ack / silence	✗	
Receiver severity/namespace/cluster filters	● dead	columns never read
Settings-based routing	🗑	migrated by migrate_alert_routing.py

1.6 Application / service management

Feature	Status	Evidence
App Services CRUD, workloads, live health	✓	ApplicationServicesPage.jsx → application_service_service.py
App Service topology + DR counterpart	✓	
Topology Components with HTTP/TCP/heartbeat health	✓	ComponentsPage.jsx → topology_component_service.py
Clients CRUD + links	✓	ClientsPage.jsx → client_service.py
Client connectivity overlay + access topology	✓	client_service_connection_service.py
Egress connections	🗑	ClientServiceEgressConnection DEPRECATED
Service Blueprints CRUD	✓ older gen	BlueprintsPage.jsx → service_blueprint_service.py

Feature	Status	Evidence
Deploy from Blueprint	● record-only	blueprint_deploy_service.py writes rows only — no Kubernetes objects created
Blueprint instances /api/app-services	●	no UI
Inventory app catalog	●	backend works; list UI disconnected (InventoryHubTable.jsx orphaned)
ApplicationDetailsPage	● hidden	deep link only

1.7 Deployment

Feature	Status	Evidence
Deploy wizard from scratch	✓	/api/inventory/deploy/wizard/* → wizard_manifest_generator.py → deployment_service.apply_yaml
Kinds: Deployment, StatefulSet, DaemonSet, Job, CronJob, Service, Ingress+TLS, ConfigMap, Secret, PVC, PV, HPA	✓	manual PV blocked for non-admins (●)
Env, secretKeyRef/configMapKeyRef, probes, volumes, nodeSelector, imagePullSecrets, requests/limits	✓	
Tolerations, affinity, initContainers, securityContext, serviceAccountName, hostAliases	✗	
StatefulSet headless Service	●	serviceName set, Service not emitted
Ingress class	●	no ingressClassName; Builder NGINX class not default. Uncertain
Namespace select / create	✓	
User template marketplace + YAML import	✓	user_template_service.py; built-ins retired
Schema-driven template deploy	✓	template_resolver.py
Raw YAML validate/dry-run/diff/apply	✓	
Deploy from Docker image	●	AddAppModal shown only to users without apps:deploy
Deploy from source	✗	only via ticket automation
Registry connections + image check (off/warn/block)	✓	registry_client.py; not applied to Helm
Deployment approval gate	●	apply_yaml only
Restart / scale / rollback / rollout history	✓	Deployments only
Rollout watch + auto-undo	✓	bundles & automation only
Deployment versions (snapshot, compare, rollback)	✓	wizard applies only
Helm releases, install, template, dry-run, rollback, uninstall	✓	helm_service.py (real binary + confirmation phrase)
Helm upgrade, repos, chart search	●	no UI
Helm chart templates (YAML, Git, archive, live namespace; versions)	✓	helm_chart_template_service.py, helm_namespace_import.py
Excel deployment forms	✓	deployment_form_service.py; bundle/approval ●
Deployment requests (quorum, signed email links, auto-decline)	✓	deployment_request_service.py
Change bundles (cart, diff, window, quorum, scheduled exec, rollout watch, emails)	✓	change_bundle_service.py, change_bundle_executor.py
Bundle action edit_resource	● broken edge	type rejected by backend

1.8 CI/CD (native KubeSight CI)

Feature	Status	Evidence
Register CI service	✓	RegisterServiceWizard.jsx → POST /api/ci/services → ci/catalog.py
Repository URL	✓ Bitbucket Cloud only	GitLab / GitHub ✗
Repository credentials	●	CI UI forces read_only=True ; merge-check delivery needs write; update/delete ●
Test source, branches/tags, revision preview	✓	
App types (container, Maven, Gradle, JAR, node, python, android, ios, flutter, generic)	✓	
Framework / build-system versions (profile)	●	JSON profile; only app type drives execution
Manual stage editor	✓	PipelineEditor.jsx → ci/pipelines.py
Starter templates	✓	ci/templates.py
Minimal default pipeline generated at build time	✓	ci/default_pipelines.py (Bitbucket API probes, no clone)
Hermes repo analysis + pipeline generation with repair loop	✓ needs Hermes	ci_assist/*, ci/generated.py
Generated pipeline review / accept	●	bypassed by auto-accept
Stage types checkout / command / container_image	✓	
Stage types publish_artifact / scan	✗	always skipped
Parallel stages	✗	parallel_group stored, never read
when conditions, parameters, variables	✓	
Secrets + log masking	✓	
Host aliases per stage	✓	pod hostAliases, BuildKit add-hosts, agent --add-host
Engine, ticker, DB queue (SKIP LOCKED), concurrency	✓	ci/engine.py, ci/ticker.py, ci/queue.py
Runner matching	●	first stage picks runner for whole build; labels ignored
Kubernetes runner	✓ off by default	ci/runners/kubernetes.py
Linux / macOS pull agent	✓	agent/kubesight-agent.py
ssh_linux runner, per-stage PVC workspace	✗	
Workspace browser	✓	
Artifacts (local store, tickets, retention)	✓	/artifacts/purge without servid → NameError
S3 / Nexus-raw artifact store	✗	
Gradle / Maven / npm / yarn / pnpm	✓	generated shell
Rootless BuildKit image build	✓ needs infra	k8s/ci-buildkitd.yaml
Image push to Nexus + digest	✓	
Trivy scan gate	✓ needs image	Dockerfile.ci-imagetools
Build caches (PV/PVC, shared tools, node_modules)	✓ needs infra	ci/cache.py, ci/cache_layout.py
Build resources	✓	ci/resources.py
History, stage matrix, logs (polling), cancel, retry, timeouts	✓	60-min total-runtime reaper
Jenkinsfile import	✓	ci/jenkinsfile.py (2,055 lines)
Bitbucket PR merge checks (ESLint, Semgrep, Sonar, Dependency-Check/NVD → gate → status + comment)	✓ with caveats	ci/merge_checks/*; only ?secret= works with Bitbucket's HMAC. Uncertain

Feature	Status	Evidence
Build status for normal builds, push/tag triggers	✗	
Jenkins integration	🗑 fallback	jenkins_client.py
Root Jenkinsfile	🗑	old self-build; missing manifests
bitbucket-pipelines.yml	🟡 current self-CI	
Pipeline jenkins	🗑	empty file

1.9 Hermes AI

Feature	Status	Evidence
MCP server /api/mcp (JSON-RPC, per-permission list/call, audited)	✓	routes/mcp.py, mcp/tools/* — 93 tools (66 read / 27 write) in 7 domains
Application Intelligence (Quick/Deep, strict schema, Semgrep/Trivy/Syft/Hadolint, SBOMs, K8s Job or local Docker)	✓	ApplicationIntelligencePage.jsx, application_intelligence_service.py, application_worker.py, application_analysis_jobs.py
Build Verified mode	✓	application_build_verifier.py
Finding workflow + comparison	✓	
Guarded Bitbucket PR from patches	✓ uncertain	application_pull_request_worker.py
Runtime intelligence	✓	deterministic; NetworkPolicy download only
CI Assist	✓	see 1.8
Ticket agent (Hermes via MCP, confidence gate, Telegram + UI approvals)	✓ master only, uncertain	services/ticket_agent/*, mcp/tools/tickets.py; local Hermes config is no_mcp
Legacy /api/ai/*	🗑 🟡	consumer cron broken; /snapshot TypeError; /health unauthenticated
hermes-agent service account	🟡	label only; applications:execute never checked
Model selection	🟡 env	HERMES_APPLICATION_MODEL, CI_ASSIST_HERMES_MODEL, TICKET_AGENT_HERMES_MODEL
In-app AI chat	✗	chat only via Hermes CLI/Telegram
Production restriction	✗	
Hermes creating tickets	✗	reads/comments/status only
Hermes infra (Dockerfiles, compose, connect script, skill)	🟡	connect script stale + leaks token

Hermes capability table (R read · An analyse · M modify · X execute)

Capability	R	An	M	X	Surface
Cluster health / overview / nodes	✓	LLM	–	–	MCP + legacy /api/ai
Alerts	✓	LLM	toggle	–	MCP
Pod logs	✓	LLM	–	–	MCP
CI build logs + failure diagnosis	✓	✓	–	–	MCP
Deployment / inventory / events / topology	✓	–	–	–	MCP
Restart / scale / rollback / pod exec	✓	–	✓	✓	MCP
Deploy apply / dry-run / diff	✓	–	✓	✓	MCP
Create deployment request	–	–	✓	–	MCP
Helm upgrade / rollback / uninstall	✓	–	✓	✓	MCP
App Services / clients / impact	✓	✓	–	–	MCP
Repository tree / files / revisions	✓	–	–	–	MCP
Source / dependency / security analysis	✓	✓	–	–	UI + MCP read

Capability	R	An	M	X	Surface
Dockerfile	✓	✓	✓	–	UI + MCP
Patch → Bitbucket PR	–	–	✓	✓	UI only
Pipeline creation	✓	✓	✓	–	UI (CI Assist)
Pipeline editing	✓	–	✓	–	MCP
Pipeline execution	✓	–	–	✓	MCP + UI
Merge-check policy	✓	–	✓	–	MCP
Tickets get / comment / status	✓	✓	✓	–	MCP (master)
Ticket-driven deploy	✓	✓	✓	✓	MCP (validator bypassable via <code>automation_run_start</code>)
Approvals	–	–	–	human	UI + Telegram
Autonomous behaviour	–	–	–	✓	ticket agent only

MCP tools by domain (93): ci 28 · clusters 10 · workloads 8 · deploys 14 · observability 7 · apps 8 · platform 13 · tickets 5.

1.10 Ticketing & deploy automation

Feature	Status	Evidence
Provider layer + <code>/api/ticketing/<p>/*</code>	✓	<code>services/ticketing/__init__.py</code> , <code>routes/ticketing.py</code> , <code>TicketingPage.jsx</code>
Zoho client + option sync	✓ live-verified 2026-07-06	<code>zoho_sync_service.py</code>
Jira Cloud/DC parity	✓ uncertain	<code>jira_sync_service.py</code>
Inbound tickets	✓ webhook only	no polling
Ticket write-back	✓	
Field / layout editor	✓	<code>components/zoho/*</code> (shared)
Zoho whole-layout writes	● gated off	<code>ZOHO_LAYOUT_WRITE_ENABLED</code>
Internal ticketing	✗	
Deploy automation (image/env/restart; native CI first, Jenkins fallback; bundle or kubectl; rollout watch; rollback; write-back)	✓	<code>deploy_automation_service.py</code>
Queued runs, per-cluster auto-run, custom envs, job overrides	✓	settings stored on <code>JenkinsConnection</code>
Outcome emails	●	rollout failure only
Legacy <code>/api/zoho/*</code>	🗑	only <code>/inbound</code> live

1.11 Mobile applications

Feature	Status	Evidence
Register app, builds, download	✓	<code>MobileAppsPage.jsx</code> → <code>mobile_app_service.py</code>
Binaries from native CI artifacts	✓	<code>ingest_ci_build</code>
Binaries from Jenkins	🗑 fallback	
Direct upload	✓	
Storage	✓ local FS only	<code>MOBILE_ARTIFACT_DIR</code>
Signature detection + publish gate	✓	<code>binary_signature.py</code>
Re-signing	✓ Jenkins only	<code>_RESIGN_EXECUTORS = ("jenkins",)</code>
Google Play publish	✓ unverified	<code>google_play_client.py</code>
App Store Connect publish	✓ unverified	<code>app_store_client.py</code> — "review" = TestFlight Beta Review
Signing in native CI / macOS runner	●	templates + agent; not proven

1.12 Security scanning

Feature	Status
Trivy image gate in CI	✓ (infra needed)
Trivy / Semgrep / Syft / Hadolint in App Intelligence	✓
ESLint / Semgrep / Sonar CLI / Dependency-Check + NVD key (merge checks)	✓
Direct NVD API client	✗
Dockerfile hardening advice	✓
NetworkPolicy isolation for CI & analysis jobs	✓
NetworkPolicy for user apps	✗ (download only)

1.13 Integrations (actual code only)

Integration	Status	Code
Kubernetes (kubectl)	✓	<code>k8s_provider.py</code>
Helm binary	✓	<code>helm_service.py</code>
Bitbucket Cloud REST	✓	<code>application_intelligence_bitbucket.py</code> , <code>ci/source</code>
Git (clone in CI pods / workers)	✓	
GitLab / GitHub	✗	
Docker (local analysis)	✓	
BuildKit	✓	
Nexus / Docker V2 registries	✓	<code>registry_client.py</code>
Harbor / Docker Hub specific	✗	generic V2 path likely works
Jenkins	✓ legacy fallback	<code>jenkins_client.py</code>
Zoho Desk	✓	<code>zoho_client.py</code>
Jira	✓ unverified	<code>jira_client.py</code>
Hermes (OpenAI-compatible)	✓	
OpenAI / Anthropic SDK directly	✗	via Hermes only
Telegram	✓ ticket agent only	<code>ticket_agent/telegram.py</code>
Slack / generic webhooks / SMTP	✓	
WhatsApp	✗	
Prometheus	✗	
metrics-server	✓	
vSphere REST	✓ read-only	<code>vsphere_client.py</code>
Google Play / App Store Connect	✓ unverified	
Sonar	● CLI stage	
NVD	● via Dependency-Check	
Trivy / Semgrep / Syft / Hadolint	✓	

2. End-to-end workflows

Register an existing cluster. ClusterManagementPage → kubeconfig upload/paste or host/port + token/cert → `POST /api/clusters/custom` → kubeconfig built & validated → `Cluster` row + file `data/kubeconfigs/cluster-N.yaml` → immediate connection test stored → appears as `custom-N` → all resource calls run `kubectl --kubeconfig`.

Provision a cluster. Sources (vSphere, SSH credential/profile, build profile) → wizard (version, topology, VIP/LB, CNI, add-ons, VMs/IPs, workloads) → draft → `/preflight` → `/start` → executor: `base_prep` → HAProxy/Keepalived → image pulls → `kubeadm`

init → CNI → CP joins → worker joins → verify → auto-onboard → add-ons → workloads → completed → download kubernetes / open in Clusters. Day two: add workers, copy workloads.

Deploy an application. Inventory → template or from scratch → eligibility → validate name → generate → prerequisites → server dry-run + image check → diff → apply (RBAC/ACL → approval gate → block-mode image check → ensure namespace → `kubectl apply`) → audit + catalog entry + version snapshot. Or "Add to bundle" → quorum → scheduled execution → rollout watch/undo.

Deployment request. ClustersPage → request → signed-link emails → votes (UI or email) → quorum → requester's applies allowed in the window → undecided requests auto-declined.

Register a CI service. Service Catalog → wizard → create service (empty default pipeline + starter Dockerfile) → Bitbucket source + credential → Hermes mode: REST evidence → proposal → validate/repair → pipeline + profile auto-saved; manual mode → Pipeline tab.

Create a pipeline. Template, Jenkinsfile import (draft) or manual stages → `PUT /pipelines/<id>`. No stages → default generated at build time.

Run a CI build. RunBuildModal → `POST /services/<id>/builds` → snapshot + queued → ticker claims (SKIP LOCKED) → runner chosen from first stage → Kubernetes Job (Secret, NetworkPolicy, init-container stages: checkout → commands → buildctl/Trivy/crane → collector) → masked logs pumped → artifacts + digest via `/api/ci/worker/*` → finalised → UI polls. Agent path: heartbeat → claim → run → stream logs → result.

Jenkinsfile import. Paste or read from repo → `POST /pipelines/import/jenkinsfile` → parameters, env, `when`, agent labels, withCredentials → secret refs, parallel flattened → draft → user saves.

Merge check. Configure tools/events/branches/gate → `merge_check` pipeline generated → Bitbucket webhook `.../inbound/<slug>?secret=` → PR event → build + INPROGRESS status → sentinel-line metrics → gate verdict → commit status `KUBESIGHT-MERGE` + PR comment; branch restriction blocks the merge.

Repository connection. Encrypted credential profile → test source → branches/tags → used by CI checkout, CI Assist, App Intelligence and PR worker.

Hermes interaction. (a) Operator mints `ksa_` token → Hermes `mcp_servers.kubesight` → chat calls `/api/mcp` as the token owner. (b) Backend calls Hermes for App Intelligence, CI Assist, ticket agent. (c) Analysis: UI → K8s Job (checkout → build verify → scanners → Hermes) → callbacks → findings/SBOMs/topology → optional PR job.

Alert handling. Tick → due policies evaluated → `AlertHistory` active → receivers/groups (email/Slack/webhook, repeat interval) → delivery log → metric auto-resolve (no notification) → UI feed/drawer/history.

Ticket → deploy. Webhook → target resolved → `ZohoInboundTicket` → (master, agent on) Hermes via MCP with confidence + approvals, else auto-run/operator → check image → build via native CI or Jenkins → verify image → bundle or `kubect`l → rollout watch → rollback on failure → outcome written back.

Mobile build & publish. Ticket → CI build → artifact ingest (or Jenkins / upload) → signature probe → Jenkins re-sign if unsigned → admin publish → Google Play edit/commit or App Store upload + processing poll → ticket comment + admin email.

3. Hidden / disconnected features

Item	Kind	Detail
ApplicationDetailsPage	no entry point	deep link only
Inventory list (<code>InventoryHubTable.jsx</code>)	orphaned	data loaded but never rendered
AddAppModal (image / YAML / repo Helm)	gated backwards	only for users without <code>apps:deploy</code>
ImageRegistriesPage	hidden nav	config moved to Settings → Integrations
<code>AlertRoutingPage.jsx</code> default export	never mounted	tabs reused
AlertRoutingModal, HealthRing, LiveLogViewer	unimported	

Item	Kind	Detail
Dashboard widget registry (~20 widgets)	dead	only OpsDashboard + myAccess render
TicketAgentConfigPanel.jsx	possibly no importer	uncertain
56 exported API functions	unused wrappers	e.g. listCiBuilds, upgradeHelmRelease, listHelmRepos, addImportToBundle, listAppServices, getBuildWorkloadPlan, update* profile functions, replaceUserAccessRules, validateCiGeneratedPipeline
Backend-only endpoints	●	/api/app-services, Helm status/releases/repos/charts/upgrade, form bundle/approval, /api/inventory/catalog/:id, /api/users/:id/access-rules*, PUT /roles/:id/permissions, workload-plan, /api/ci/builds, alert evaluate/stats
Machine-to-machine endpoints	by design	/api/mcp, /api/ci/agent/*, /api/ci/worker/*, analysis/PR callbacks, webhooks, heartbeat, email links
Legacy /api/ai/*, /api/zoho/*	no UI	
Unused facades	dead	alert_service.py, cluster_service.py, kubernetes_service.py, settings_service.py
Unused models	dead	AlertNotificationSent, ClientServiceEgressConnection
Unused columns	dead	AlertHistory.last_notified_at, receiver filters, builder ipam_* / lb_config_json, version ci_*, CI parallel_group, runner labels, CI blueprint_id / catalog_entry_id
Feature flags		KUBESIGHT_ENABLE_CILIUM, ZOHO_LAYOUT_WRITE_ENABLED, CI_ASSIST_POLICY, K8S_REAL_MODE, AUTH_REQUIRED, CI_ENGINE_TICKER, ALERT_POLICY_SCHEDULER, CACHE_WARM_ENABLED, CI_ARTIFACT_AUTOCLEAN

4. Duplicate / legacy systems

Domain	Current	Legacy / older	Verdict
Service catalog	CiService at #/service-catalog	ServiceBlueprint + AppService (#/blueprints, deploy records only); AppCatalogEntry; IntelligenceApplication duplicates repo fields	four overlapping registration models
"App service" naming	ApplicationService	AppService (mirrored)	confusing twins
CI	native CI	Jenkins fallback, root Jenkinsfile, empty Pipeline jenkins	Jenkins removal (phases 6–7) not done
Self-CI	bitbucket-pipeline s.yml	Jenkinsfile	Jenkinsfile legacy
Ticketing	/api/ticketing/<p>	/api/zoho/*	components/zoho/* is current shared UI
Provisioning	cluster_build	none; upgrade_executor duplicates OS scripts; custom-resources.yaml unused	
Manifest generation	wizard_manifest_generator.py	manifest_generator.py	both live
Wizard templates	user templates	built-ins (TEMPLATES = [])	retired
Logs API	cluster logs + SSE	GET /api/logs	legacy, used once
Access grants	AccessRules	legacy tables	fallback
Alert routing	receivers & groups	settings routing	migrated
AI access	MCP	/api/ai/*	legacy
Dashboard	OpsDashboard	widget registry	dead code shipped

Domain	Current	Legacy / older	Verdict
Frontend API	frontend/src/api/*	api.js barrel	legacy, still imported by 17 files
Design system	Signal sg- classes + Cardinal tokens	premium.css , ui-polish.css ; stale MASTER.md	docs drifted
Migrations	migrate_rbac.py	Alembic	dead

5. Database architecture

Model	Purpose	Relationships	Feature
User	account, onboarding, MFA, lockout	→ Role, AccessRule	Auth
Role / Permission	RBAC	many-to-many	RBAC
AccessRule	scoped allow/deny	→ User	RBAC
ApiToken	hashed token	→ User	MCP
AuditLog	audit trail	actor → User	Audit
AppSettings	workspace settings	–	Settings
Cluster	registered cluster	string id custom-N	Clusters
AlertPolicy / AlertHistory / LogAlertSeen	rules, instances, dedup	policy ↔ receivers/groups	Alerts
AlertRouting Smtip / Receiver / ReceiverGroup / DeliverySent / DeliveryLog	routing		Alerts
ApplicationService (+ Deployment, TopologyNode/Edge)	curated app service	↔ Client	App Services
TopologyComponent	infra block + health	← nodes	Components
Client / ClientApplicationService / ClientServiceConnection	customers & connectivity	→ ApplicationService	Clients
ServiceBlueprint (+ Component, Connection, Requirement)	architecture template	1-N AppService	Blueprints
AppService / AppServiceComponentMapping	blueprint instance	→ blueprint, client, mirror	Blueprint deploy
AppCatalogEntry / ApplicationDeploymentVersion	inventory metadata, snapshots	1-N	Inventory
UserTemplate / HelmChartTemplate (+ Version)	templates, charts		Deploy, Helm
DeploymentRequest / Vote / Setting	quorum approval		Requests
ChangeBundle / Item / Vote / BundleRolloutWatch	staged changes		Bundles
DeploymentFormGeneration / Import	Excel forms		Forms
RegistryConnection	registry profile	← CiService, CiArtifact	Registry
CiService	CI catalog entry	→ credential, registry; owns pipelines, builds, secrets, analyses	CI
CiPipeline / CiPipelineStage	pipeline definition	→ service	CI
CiBuild / CiBuildStage / CiLogChunk	runs & logs	→ runner	CI
CiRunner / CiAgentTask	runners, pull tasks		CI
CiArtifact / CiSecret	outputs, secrets		CI
CiRepositoryAnalysis	Hermes proposal	→ service	CI Assist

Model	Purpose	Relationships	Feature
CiMergeCheckPolicy / Config / CiMergeCheck	gate, webhook, PR evaluation	→ service, build	Merge checks
BitbucketCredentialProfile	shared repo credential	← CiService, IntelligenceApplication	CI, App Intel
IntelligenceApplication / ApplicationAnalysis / Finding / Dependency / Communication / Artifact / RuntimeSnapshot / PullRequest	Hermes analysis		App Intelligence
VSphereConnection / SshCredential / SshConnectionProfile / SshHostKey / BuildProfile	build sources		Builder
ClusterBuild / ClusterBuildNode / ClusterBuildStep	build, nodes, phases	→ Cluster	Builder
ZohoIntegration / JiraIntegration / TicketingDeployConfig	provider config		Ticketing
ZohoInboundTicket / FieldBinding / LayoutSnapshot / DeploymentSnapshot	tickets & bindings (both providers)		Ticketing
JenkinsConnection / DeployAutomationRun	Jenkins + auto-run settings; runs	→ CiBuild, ChangeBundle	Automation
MobileApplication / Build / Resign / Publish	mobile apps	→ CiService	Mobile
TicketAgentSettings / TicketInterpretation (master)	agent config & tasks	→ ticket, run	Ticket agent
AlertNotificationSent, ClientServiceEgressConnection	unused		Legacy

6. API inventory (summarised)

- **Auth** `/api/auth` : login, mfa/verify, first-login change-password / totp setup / totp verify, me, logout, tokens (GET/POST/DELETE).
- **Users & roles** `/api/users` CRUD + permanent delete + unlock / reset-mfa / resend-temp / force-reset / lock / enable; `/api/roles` CRUD; `/api/permissions` .
- **Audit & settings** : `/api/audit-logs` (+ `/export`); `/api/settings` .
- **Clusters** `/api/clusters` : list; custom CRUD + test; overview, nodes, storageclasses, namespaces, pod-issues, topology; namespace resources / events / topology / pods / config-resources; describe, yaml, restart, exec; container logs + SSE stream; rollout history.
- **Dashboard & logs** : `/api/dashboard/summary` ; `/api/logs` (legacy).
- **Alerts** : `/api/alerts` ; `/api/alert-policies` CRUD + catalog / history / stats / evaluate / status; `/api/alert-routing/{smtp, receivers, receiver-groups, delivery-logs}` .
- **Upgrades** : `/api/upgrades/info` , `/precheck` , `/start` , `/jobs/:id` .
- **Cluster Builder** : `/api/cluster-builds` options, CRUD, preflight/start/retry/cancel, logs, nodes, grow-preflight, grow, workloads, workload-sources, workload-plan, bring-workloads, kubeconfig; infra `/api/{vsphere-connections, ssh-credentials, ssh-connection-profiles, build-profiles}` CRUD + test + vms.
- **App services / clients / blueprints** : `/api/application-services` CRUD + pickers; `/api/topology-components` CRUD + check + heartbeat; `/api/clients` CRUD + services + connection + topology; `/api/service-blueprints` CRUD + deploy-plan + deploy + pickers; `/api/app-services` .
- **Inventory & deploy** `/api/inventory` : catalog CRUD, register, workloads, versions (compare / rollback), actions (restart / scale / rollback / rollout-history), deploy/yaml/, `deploy/image/`, `deploy/wizard/*` + templates.
- **Helm** `/api/helm` : status, releases, repos, charts, template, dry-run, install, upgrade, rollback, uninstall, confirmation-phrase, chart-templates/* (yaml, git, archive, namespace, versions).
- **Registries** : `/api/registries` CRUD + test + check-image.
- **Forms / requests / bundles** : `/api/deployment-forms/*` ; `/api/deployment-requests/*` (incl. GET email approve/decline); `/api/change-bundles/*` (draft, items, diff, submit, approve, reject, email links).
- **CI** `/api/ci` : services, source (test / branches / revisions / credentials / file), pipelines (templates, from-template, import/jenkinsfile, lint, portability), builds (run, cancel, retry, workspace, stage logs + download), artifacts (tickets, purge), secrets, runners (+ agents, token), cache, assist (availability, capabilities, analysis, accept, cancel, profile, validate-generated),

merge-checks (policy, service config, secret, enforcement, runs, redeliver, inbound), worker (meta, artifacts), agent (heartbeat, claim, tasks logs / artifacts / result).

- **Application Intelligence:** `/api/applications` CRUD + analyses; `/api/application-analyses/:id/*` (compare, cancel, findings, topology, apis, configuration, runtime, artifacts, pull-requests); `/api/bitbucket-credential-profiles`; `hermes/test; bitbucket/*`; worker callbacks.
- **Hermes / AI:** `/api/mcp` (POST JSON-RPC, GET discovery); `/api/ai/*` (legacy); `/api/ticket-agent/{settings, telegram/test, tasks/:id/approve|reject, tickets/:id/handle}` (master).
- **Ticketing:** `/api/ticketing/providers`; `/api/ticketing/:p/{config, test, sync, preview, source/*, layout, fields/*, sections, option-sources, layout/snapshots/*, inbound-tickets, jenkins, automation/runs, inbound}`; `/api/zoho/*` (legacy).
- **Mobile** `/api/mobile-apps` : CRUD, environments, test-jenkins / test-play / test-appstore, fetch, builds upload / list / download / delete, resign, publish (admin).
- **Integrations:** `/api/integrations`, `/:key`, `/:key/test`, `/:key/enabled`, `/:key/activity`.

7. Capability matrix

Area	Feature	Status	FE	BE	E2E	Notes
Core	Login + TOTP MFA + onboarding	✓	Yes	Yes	Yes	TOTP secret plaintext
Core	Lockout / admin unlock	✓	Yes	Yes	Yes	
Core	Self-service password / MFA	✗	Display	No	No	
Core	Logout revocation	●	Yes	Audit	No	
Core	Users CRUD	✓	Yes	Yes	Yes	
Core	Roles & permissions	✓	Yes	Yes	Yes	
Core	Scoped access rules	✓	Yes	Yes	Yes	
Core	API tokens	●	Yes	Yes	Yes	no perm check / scopes
Core	Audit log + CSV	✓	Yes	Yes	Yes	
Core	Settings	✓	Yes	Yes	Yes	
Core	Secret encryption	✓	–	Yes	Yes	not kubeconfigs
Core	Mock / real mode	✓	–	Yes	Yes	
Cluster	Register cluster	✓	Yes	Yes	Yes	
Cluster	Auto-discover contexts	✓	Yes	Yes	Yes	
Cluster	Update / test cluster	✓	Yes	Yes	Yes	
Cluster	Delete cluster	●	Yes	Soft	Partial	file left
Cluster	Multi-cluster list	✓	Yes	Yes	Yes	
Cluster	Cluster health	●	Yes	Basic	Yes	
Cluster	Overview	●	Yes	Partial	Yes	zeros
Cluster	Namespaces	●	Yes	Read	Yes	
Cluster	Pods view/logs/exec/restart	✓	Yes	Yes	Yes	
Cluster	Deployments view/edit/restart	✓	Yes	Yes	Yes	
Cluster	Deployments scale/rollback	✓	Hidden page	Yes	Partial	
Cluster	StatefulSet / DaemonSet	●	Yes	View+restart	Yes	
Cluster	ReplicaSet / Job / CronJob	●	Yes	View	Yes	
Cluster	Services	●	Yes	View	Yes	
Cluster	Ingress / ConfigMap / Secret edit	✓	Yes	Yes	Yes	secret data exposed
Cluster	Events	✓	Hidden page	Yes	Partial	
Cluster	PVC / PV browse	✗	No	No	No	

Area	Feature	Status	FE	BE	E2E	Notes
Cluster	StorageClass list	🟡	Wizard	Yes	Yes	
Cluster	Nodes page / cordon / drain	❌	No	Upgrade only	No	
Cluster	Resource delete	❌	No	Bundle only	No	
Cluster	Topology map	✅	Yes	Yes	Yes	
Cluster	K8s upgrade	🟡	Yes	Yes	Uncertain	managed clouds precheck
Builder	Wizard / preflight	✅	Yes	Yes	Uncertain	real VMs pending
Builder	kubeadm phase machine	✅	Yes	Yes	Uncertain	
Builder	HA + HAProxy/Keepalived/VIP	✅	Yes	Yes	Uncertain	
Builder	Calico / Flannel	✅	Yes	Yes	Uncertain	
Builder	Cilium	🟡	Flag	Partial	No	
Builder	metrics-server / NGINX / MetalLB	✅	Yes	Yes	Uncertain	
Builder	Onboard + kubeconfig	✅	Yes	Yes	Yes	
Builder	Add workers	✅	Yes	Yes	Uncertain	CSR gap
Builder	Remove node / teardown / IPAM	❌	No	No	No	
Builder	Copy workloads + NFS plan	✅	Yes	Yes	Uncertain	
Builder	vSphere / SSH profiles	✅	Yes	Yes	Yes	edit BE only
Builder	Host-key pinning	🟡	Hint	Partial	No	
Monitor	CPU / mem	✅	Yes	Yes	Yes	
Monitor	Network metrics	🟡	Fake	No	No	
Monitor	Disk metrics	❌	No	No	No	
Monitor	Historical metrics	🟡	Buffer	No	No	
Monitor	Prometheus	❌	No	No	No	
Monitor	Dashboard	✅	Yes	Yes	Yes	
Monitor	Logs + SSE	✅	Yes	Yes	Yes	
Monitor	Aggregated logs	❌	No	No	No	
Alerts	Metric policies	✅	Yes	Yes	Yes	
Alerts	PVC / disk policies	🟡	Yes	Fake	Misleading	
Alerts	Log alerts	✅	Yes	Yes	Yes	no resolve
Alerts	Service / automation alerts	✅	Yes	Yes	Yes	
Alerts	Email / Slack / webhook	✅	Yes	Yes	Yes	
Alerts	Telegram / WhatsApp alerts	❌	No	No	No	
Alerts	Ack / silence	❌	No	No	No	
Apps	App Services + topology + DR	✅	Yes	Yes	Yes	
Apps	Topology components	✅	Yes	Yes	Yes	
Apps	Clients + connectivity	✅	Yes	Yes	Yes	
Apps	Blueprints CRUD	✅	Yes	Yes	Yes	older gen
Apps	Deploy from Blueprint	🟡	Yes	Records	No	no K8s objects
Apps	Inventory list	🟡	Orphaned	Yes	No	
Apps	Application details	🟡	Deep link	Yes	Partial	
Deploy	Wizard deploy	✅	Yes	Yes	Yes	
Deploy	Templates / schema deploy	✅	Yes	Yes	Yes	
Deploy	Raw YAML apply	✅	Yes	Yes	Yes	
Deploy	Deploy from image	🟡	Gated backwards	Yes	No	

Area	Feature	Status	FE	BE	E2E	Notes
Deploy	Deploy from source	✗	No	Tickets only	No	
Deploy	Env / secrets / probes / limits	✓	Yes	Yes	Yes	
Deploy	Services / Ingress	✓	Yes	Yes	Yes	class gap
Deploy	PVC multi-volume	✓	Yes	Yes	Yes	
Deploy	Manual PV	●	Yes	Admin	Partial	
Deploy	hostAliases for apps	✗	No	No	No	
Deploy	Registry image check	✓	Yes	Yes	Yes	not Helm
Deploy	Approval gate	●	Yes	apply_yaml	Partial	
Deploy	Scale / restart / rollback	✓	Hidden	Yes	Partial	
Deploy	Versions + rollback	✓	Hidden	Yes	Partial	
Deploy	Helm install / rollback / uninstall	✓	Yes	Yes	Yes	
Deploy	Helm upgrade / repos / search	●	No	Yes	No	
Deploy	Helm chart templates + ns import	✓	Yes	Yes	Yes	
Deploy	Excel forms	✓	Yes	Yes	Yes	
Deploy	Deployment requests	✓	Yes	Yes	Yes	
Deploy	Change bundles	✓	Yes	Yes	Yes	
CI	Register service	✓	Yes	Yes	Yes	
CI	Bitbucket source + creds	●	Yes	Yes	Yes	read-only forced
CI	GitLab / GitHub	✗	No	No	No	
CI	Manual editor	✓	Yes	Yes	Yes	
CI	Templates / default pipeline	✓	Yes	Yes	Yes	
CI	Hermes repo analysis	✓	Yes	Yes	Uncertain	
CI	Hermes pipeline generation	●	Yes	Yes	Yes	auto-accept
CI	Framework / version detection	●	Yes	Profile	Partial	
CI	Pipeline execution	✓	Yes	Yes	Uncertain	mock-runner risk
CI	Queue / concurrency	✓	Yes	Yes	Yes	
CI	Runner matching	●	Yes	First stage	Partial	
CI	Kubernetes runner	✓	Yes	Yes	Uncertain	off by default
CI	Linux / macOS agent	✓	Yes	Yes	Uncertain	
CI	ssh_linux runner	✗	No	No	No	
CI	BuildKit build + push	✓	Yes	Yes	Uncertain	infra
CI	Trivy gate	✓	Yes	Yes	Uncertain	image
CI	Gradle / Maven / npm / yarn / pnpm	✓	Yes	Yes	Uncertain	
CI	Parallel stages	✗	No	No	No	
CI	publish_artifact / scan stages	✗	–	Skipped	No	
CI	Params / secrets / masking	✓	Yes	Yes	Yes	
CI	Host aliases	✓	Yes	Yes	Yes	
CI	Artifacts (local)	✓	Yes	Yes	Yes	purge bug
CI	S3 / Nexus-raw	✗	No	No	No	
CI	Caches	✓	Yes	Yes	Uncertain	
CI	Build resources	✓	Yes	Yes	Yes	
CI	Logs / history / stage matrix	✓	Yes	Yes	Yes	
CI	Cancel / retry / timeouts	✓	Yes	Yes	Yes	60-min reaper

Area	Feature	Status	FE	BE	E2E	Notes
CI	Workspace browser	✓	Yes	Yes	Yes	
CI	Jenkinsfile import	✓	Yes	Yes	Yes	
CI	Merge checks	✓	Yes	Yes	Uncertain	webhook caveat
CI	Push / tag triggers	✗	No	No	No	
CI	Jenkins integration	🗑️	Yes	Yes	Yes	fallback
Hermes	MCP server (93 tools)	✓	Tokens	Yes	Uncertain	
Hermes	Read health / alerts / logs	✓	–	Yes	Uncertain	
Hermes	Workload ops / exec / deploy / Helm	✓	–	Yes	Uncertain	
Hermes	Execute pipeline	✓	–	Yes	Uncertain	
Hermes	Edit pipeline / Dockerfile	✓	–	Yes	Uncertain	
Hermes	App Intelligence	✓	Yes	Yes	Uncertain	
Hermes	Security / deps / SBOM	✓	Yes	Yes	Uncertain	
Hermes	Patch → PR	✓	Yes	Yes	Uncertain	
Hermes	CI Assist generation	✓	Yes	Yes	Uncertain	
Hermes	Ticket agent + Telegram	✓	Yes	Yes	Uncertain	master only
Hermes	Ticket creation	✗	No	No	No	
Hermes	In-app chat	✗	No	No	No	
Hermes	Service account enforcement	🟡	–	Label	No	
Hermes	Production restriction	✗	No	No	No	
Hermes	Legacy /api/ai	🗑️	No	Yes	Broken	
Ticketing	Zoho config / sync / editor	✓	Yes	Yes	Yes	
Ticketing	Jira parity	✓	Yes	Yes	Uncertain	
Ticketing	Inbound webhook	✓	Yes	Yes	Yes	open w/o secret
Ticketing	Write-back	✓	–	Yes	Uncertain	
Ticketing	Ticket → deploy	✓	Yes	Yes	Uncertain	
Ticketing	Rollback + email	✓	–	Yes	Uncertain	
Ticketing	Zoho layout writes	🟡	Yes	Gated	No	
Ticketing	Internal ticketing	✗	No	No	No	
Mobile	Registry / builds / download	✓	Yes	Yes	Yes	
Mobile	CI ingest	✓	Yes	Yes	Uncertain	
Mobile	Jenkins fetch	🗑️	Yes	Yes	Yes	
Mobile	Local binary store	✓	–	Yes	Yes	no bucket
Mobile	Signature gate	✓	Yes	Yes	Yes	
Mobile	Re-sign (Jenkins)	✓	Yes	Yes	Uncertain	
Mobile	Google Play	✓	Yes	Yes	Uncertain	
Mobile	App Store (TestFlight)	✓	Yes	Yes	Uncertain	
Mobile	macOS CI signing	🟡	Yes	Templates	No	
Storage	PVC via wizard	✓	Yes	Yes	Yes	
Storage	PV via wizard	🟡	Yes	Admin	Partial	
Storage	NFS static PVs (Builder)	✓	Yes	Yes	Uncertain	
Storage	CI cache PV/PVC	✓	Yes	Yes	Uncertain	
Storage	StorageClass / provisioner	✗	No	No	No	
Network	ClusterIP / NodePort / LB	✓	Yes	Yes	Yes	

Area	Feature	Status	FE	BE	E2E	Notes
Network	MetalLB	✓	Yes	Yes	Uncertain	Builder
Network	HAProxy / Keepalived / VIP	✓	Yes	Yes	Uncertain	Builder
Network	Calico	✓	Yes	Yes	Uncertain	
Network	NetworkPolicy (internal jobs)	✓	–	Yes	Yes	
Network	NetworkPolicy (apps)	✗	Download	No	No	
Network	DNS config	●	No	Verify	–	
Integr.	Slack / webhook / SMTP	✓	Yes	Yes	Yes	
Integr.	Telegram	✓	Yes	Yes	Uncertain	ticket agent
Integr.	WhatsApp	✗	No	No	No	
Integr.	Nexus / V2 registry	✓	Yes	Yes	Yes	
Integr.	vSphere	✓	Yes	Yes	Yes	read-only
Integr.	Integrations hub	✓	Yes	Yes	Yes	

8. Executive summary

What KubeSight can do today

- **Multi-cluster Kubernetes console:** register clusters by kubeconfig or manually; browse 11 namespaced kinds; YAML edit/apply with dry-run and diff; restart; exec; logs with live follow; topology map; metrics-server CPU/memory dashboard.
- **Cluster Builder:** provision kubeadm clusters on existing VMs over SSH — single or HA control plane, HAProxy/Keepalived VIP, Calico/Flannel, metrics-server, NGINX, MetalLB, offline/mirror profiles, auto-onboarding, add workers, copy workloads with NFS storage.
- **Native CI:** Bitbucket service catalog; stage editor, templates and generated defaults; Kubernetes Job runner and Linux/macOS agent; rootless BuildKit builds with Trivy gating and Nexus push; caches, artifacts, secrets, parameters, host aliases; logs, retry, cancel, stage matrix; Jenkinsfile import; Bitbucket PR merge checks with quality gates.
- **Hermes AI:** MCP server with 93 RBAC-filtered tools; source analysis (findings, SBOMs, architecture, guarded fix PRs); Hermes-generated pipelines; autonomous ticket agent with Telegram approvals (master).
- **Governed deployment:** wizard + templates, Helm (incl. charts imported from a live namespace), Excel forms, registry image gates, quorum deployment requests with email approvals, scheduled change bundles with rollout watch and auto-undo.
- **Ticket-driven deploy automation:** Zoho/Jira → image/env/restart → native CI build → rollout → auto-rollback → write-back.
- **Mobile release management:** ingest APK/AAB/IPA, signature checks, Jenkins re-sign, Google Play and TestFlight publishing.
- **Alerting:** metric/log/service/automation policies with email/Slack/webhook routing and receiver groups.
- **Application landscape modelling:** App Services with topology and DR, topology components with health checks, clients with per-client connectivity.
- **Enterprise auth:** MFA onboarding, logout, roles, fine-grained access rules, audit log with CSV export, API tokens.

Partially complete

Runner matching (first stage only); CI Assist review (auto-accepted); merge-check credential write mode; Blueprint deploy (records only); approval gate (not on Helm / restart / scale / rollback); cluster delete (soft); cluster overview zeros; K8s upgrades (kubeadm/minikube, jobs in memory); Cilium; SSH host-key pinning; grow-workers CSR approval; PVC/disk alert values; log alerts never resolve; API tokens (no scopes/check); hermes-agent account (label only); Zoho layout writes (gated off); deploy-from-image (gated backwards); Ingress class and StatefulSet headless Service; macOS CI signing.

Only UI / prototype

Dashboard network chart and metric history (fake); Settings → Security card; concept/ mockups and cluster-builder-concept.html; dead dashboard widget registry.

Exists in the backend but isn't exposed

`/api/app-services`; Helm upgrade/repos/search; deployment-form → bundle/approval; access-rules sub-API; vSphere/SSH/build-profile edits; global CI builds list, portability, generated-pipeline validation; alert evaluate/stats; `/api/inventory/catalog/:id`; legacy `/api/ai/*` and `/api/zoho/*`; the MCP write surface (by design).

Legacy

Jenkins CI (fallback), root Jenkinsfile, empty `Pipeline jenkins`, Alembic; `/api/ai`, `/api/zoho`, `GET /api/logs`; legacy access grants and settings-based alert routing; `AlertNotificationSent`, `ClientServiceEgressConnection`; `api.js` barrel; `premium.css` / `ui-polish.css`; built-in wizard templates; four unused service facades; the ServiceBlueprint/AppService generation (semi-legacy).

Planned but not implemented

Prometheus, disk/network metrics, stored history; PVC/PV browsing and resource delete; alert ack/silence and Telegram/WhatsApp alerts; parallel stages; `ssh_linux` runner; publish/scan stages; S3 artifacts; GitLab/GitHub; push/tag CI triggers; build status for normal builds; per-stage PVC workspaces; IPAM, kube-vip; live node removal; teardown; StorageClass/NFS provisioner add-on; hostAliases and NetworkPolicy for apps; deploy-from-source; internal ticketing; Hermes ticket creation; in-app AI chat; production restrictions; token scopes; full Jenkins removal.

The 10 largest functional areas (by scope of code)

1. Native CI platform — engine, runners, BuildKit, caches, merge checks, Jenkinsfile import.
2. Cluster Builder — kubeadm provisioning, HA, add-ons, day two.
3. Deployment & governance — wizard, templates, Helm catalog, forms, requests, change bundles.
4. Kubernetes resource console — clusters, resources, logs, topology.
5. Application Intelligence — Hermes source analysis, scanners, PR generation.
6. Ticketing & deploy automation — Zoho/Jira, field sync, ticket → build → deploy.
7. Alerting & notification routing.
8. Hermes MCP server & ticket agent — 93 tools.
9. Identity & access — MFA onboarding, RBAC, scoped access rules.
10. Mobile application release — ingest, signatures, re-signing, Play/App Store publishing.